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UNO Q Based Networking: LAN, Stack and Cyber Security Study Trainer (Vi LaN-05)

Introduction

The **UNO Q Based Network LAN Trainer with Network Stack and Cyber Security Studies** is an advanced embedded networking platform developed for learning and experimentation

MAKE : VI MICROSYSTEMS

MODEL: ViLaN-05

- Ethernet communication
- TCP/IP network stack implementation
- LAN technologies
- Socket programming
- IoT networking
- Network protocol analysis
- Cyber security techniques
- Secure embedded communication

The trainer is built around the Arduino UNO Q and integrates Ethernet, serial communication, wireless networking, and cyber security modules for practical hands-on learning.

An indigenous design of LAN Trainer for wired & wireless to study for all topologies like **star, bus, mesh and ring** has been introduced. All protocol ssuch as **Stop & Wait, Go Back to N, Selective Repeat, Token Ring, Token Bus, CSMA / CD, CSMA / CA can be experimented using Vi LaN-05**. A Microcontroller with 2 Ethernet ports forms one Station. There are 4 such stations and one station connected to a PC for Graphical Representation of all experiments.

All topologies are implemented using 10/100Mbps Ethernet ports for real feel of Networking. Vi LaN-05 can conduct all experiments as per Anna University Syllabus of ECE Department. Data transmission speed – **10/100 Mbps**. Packet Size: 128,512,1024, 2048,4096,8192,16384, Inter packet delay: 1000-5000ms, Acknowledgement lost, bad packet, Error Generation & Error Correction.

Objectives

- Study the TCP/IP protocol stack
- Learn embedded LAN communication
- Understand Ethernet frame transmission
- Implement TCP and UDP socket communication
- Analyze network packets
- Study cyber security vulnerabilities
- Learn secure communication techniques
- Develop embedded IoT applications
- Perform real-time network monitoring

To Implement

- Networking
- Cyber Security
- Topologies
- Protocols
- Concept of Token
- Collision Detection
- Collision Avoidance
- TCP/IP
- UDP
- Ethernet II

IEEE Standards

- ◆ 802.11 Standard
 - From 802.11b to newer 802.11 versions

Features

Network Stack Studies

- Ethernet Layer implementation
- ARP protocol study
- IP addressing
- ICMP protocol
- TCP communication
- UDP communication
- DHCP configuration
- DNS operation
- Socket programming
- Client-server networking

Embedded Communication Interfaces

- UART communication
- SPI communication
- I2C communication
- USB-to-UART interface
- RS232 communication
- Parallel port interfacing

Cyber Security Studies

- Port scanning experiments
- Packet sniffing
- MAC filtering
- Authentication systems
- Password-based access
- Embedded firewall concepts
- Secure data transmission
- Encryption and decryption
- Digital signature techniques
- Network attack demonstrations

IoT and Wireless Networking

- Wi-Fi networking
- MQTT communication
- HTTP/HTTPS communication
- Cloud connectivity
- Remote embedded monitoring

Wireless Security

- WPA / TKIP
- WPA2 / AES
- WPA3 / SAE

Network Concepts

- Network Services
- MITM (Man-In-The-Middle) Attack

Cryptography

- ★ Various Algorithms
- ★ Cipher Techniques

LIST OF EXPERIMENTS

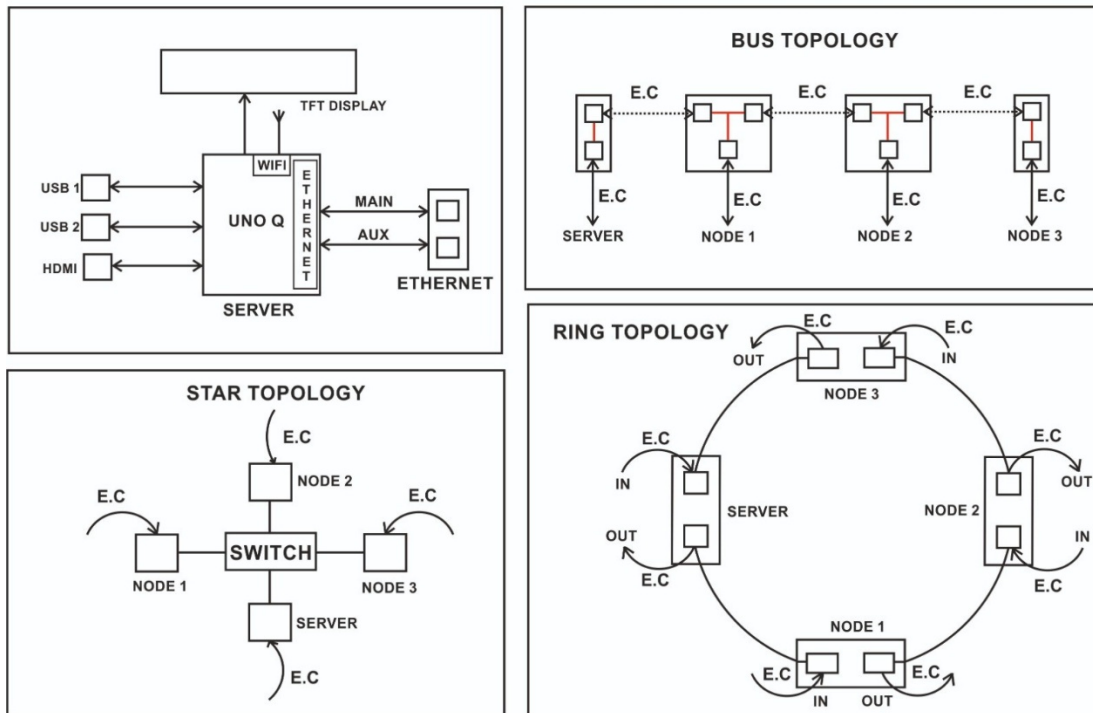
1. Topology
 - i) Star
 - ii) Bus
 - iii) Ring
 - iv) Mesh
2. Protocols Supported
 - i) Stop & Wait
 - ii) Go Back to N
 - iii) Selective Repeat
3. Token Ring
4. Token Bus
5. CSMA / CD
6. CSMA / CA
7. Serial to Serial communication
8. Parallel to Parallel Communication
9. Ethernet communication using RJ 45
10. Study of Socket Programming and Client – Server model
11. Write a code simulating ARP / RARP protocols.
12. Write a code simulating PING and TRACEROUTE commands
13. Create a socket for HTTP for web page upload and download.
14. Write a program to implement RPC (Remote Procedure Call)
15. Implementation of Subnetting.
16. Applications using TCP Sockets like
 - a. Echo client and echo server
 - b. Chat
 - c. File Transfer
17. Applications using TCP and UDP Sockets like
 - a. DNS
 - b. SNMP
 - c. File Transfer
18. Study of Network simulator (NS), and Simulation of Congestion Control Algorithms using NS
19. Perform a case study about the different routing algorithms to select the network path with its optimum and economical during data transfer.
 - a. Link State routing

- b. Flooding
 - c. Distance vector
20. Write a program to implement the following cipher techniques to perform encryption and decryption
 - i. Caesar Cipher
 - ii. Playfair Cipher
 - iii. Hill Cipher
 21. Write a program to implement the following transposition techniques
 - (i) Rail fence technique - Row major transformation
 - (ii) Rail fence technique - Column major transformation
 22. Write a program to implement DES algorithm
 23. Write a program to implement AES algorithm
 24. Write a program to implement RSA Encryption algorithm
 25. Write a program to implement the Diffie-Hellman Key Exchange mechanism.
Consider one of the parties as Alice and the other party as Bob.
 26. Write a program to calculate the message digest of a text using the SHA-1 algorithm.
 27. Write a program to calculate the message digest of a text using the MD-5 algorithm.
 28. Write a program to implement digital signature standard.

Compiler, Interpreter and OS

- Ⓟ GNU C Compiler set
- Ⓟ Python Interpreter
- Ⓟ Debian Linux

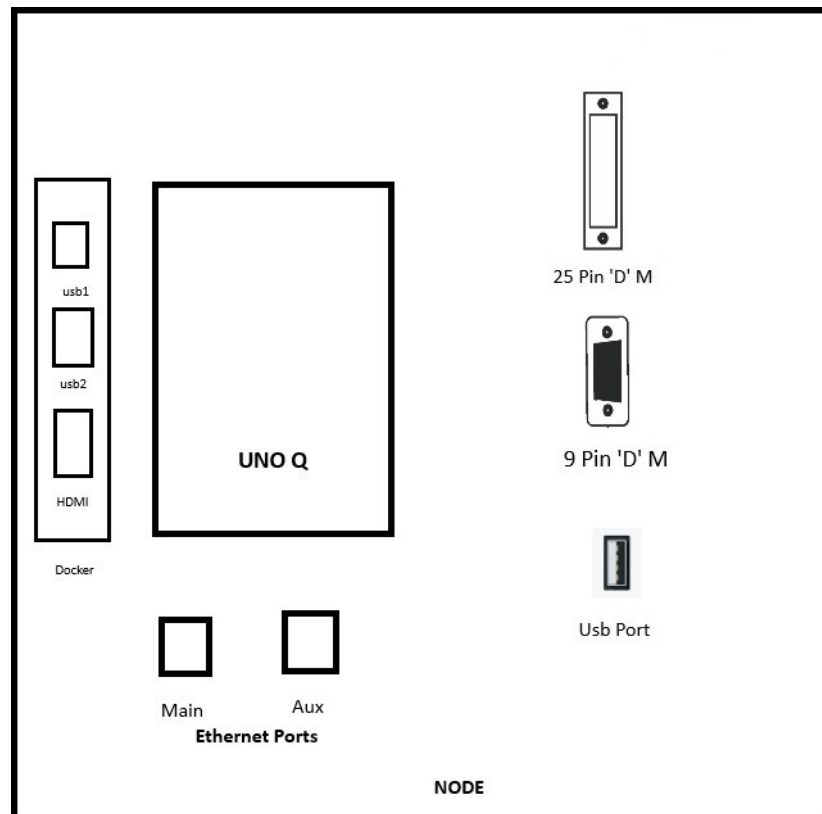
UNO Q BASED EMBEDDED NETWORK TRAINER



Features of the ViLAN-05 GUI

- Graphic representation of data packets
- Packet to be transmitted.
- Packet being transmitted.
- Packet transmitted with acknowledgment.
- Packet with acknowledgment lost error.
- Packet with bad packet error.
- Packet which is discarded or resend.
- Data Encryption & Decryption.
- Peer to Peer, Star, Bus and Ring & Mesh Topologies.
- Ethernet LAN Protocols.
- User selectable packet size.
- Manual Error generation.
- Built in Protocol, Network and graph analysis.
- Sample 'C' & .net Programs files provided for project purposes.
- Data encryption & decryption

- Complete analysis of Network & Protocols
- Graphic representation of data on s/w screen with packet details
- Network details indication of computer name, IP address, Port number, status of network, MAC address and OS on computer.
- Indication of packet serial number, file name, file size, file number, receiver name, Workgroup, receiver IP address, total packets, packet length, time out, protocol, topology, receiver, MAC address, port number, file send start time, file sent completion time, transmission time data rate (Mbps), error.



HARDWARE

- One UNO Q with 2 Ethernet Port provided in Each Node
- These 2 Ethernet ports for implementing RING Topology.
- These 2 ports are terminated using commercial D-Link Face Plate 2 Dual or 4 outlets Ethernet terminals for giving the students a commercial feel of NETWORKING
- Total of 4 Nodes provided to conduct many experiments.
- In-built Wi-Fi is provided to conduct all Wi-Fi Experiments
- 4 Port Ethernet switch provided with D-Link Face Plate 4 outlets to cable various Ethernet connections for experimentation.
- 2.4" TFT Display provided in each Node to display various parameters of Topology & Protocol in WiFi
- All components are housed in a sleek cabinet.

1. Arduino UNO Q Specifications:

Processor:

- **Qualcomm Dragonwing™ QRB2210 MPU:**
 - Processor Type: Quad-core Arm Cortex-A53
 - Clock Speed: 2.0 GHz
 - Operating System: Debian Linux
 - Applications: Python, AI workloads, Networking
 - GPU: Adreno GPU
 - ISP: Dual ISP (Image Signal Processor)
 - DSP Support: DSP acceleration
 - Supports: Camera, Display, Audio, Edge AI

MicroController:

- **STM32U585 MCU:**
 - Processor Type: Arm Cortex-M33
 - Clock Speed: Up to 160 MHz
 - Applications: GPIO control, PWM, Sensors, Deterministic timing, Low-power tasks
 - Flash Memory: 2 MB
 - SRAM: 786 KB

Memory & Storage:

- RAM: 2GB or 4GB LPDDR4
- Storage: 16GB or 32GB eMMC

- SD Card: Not required
- **Operating Systems:**
 - MPU Side: Debian Linux
 - MCU Side: Zephyr RTOS / Arduino Framework
 - System Capability: Linux + Real-time processing simultaneously
- **Wireless Connectivity:**
 - Wi-Fi: Dual-band Wi-Fi 5
 - Bluetooth: Bluetooth 5.1
 - Applications: IoT, Wireless communication,
- **Cloud applications**
 - Graphics & AI:
 - GPU: Adreno 702 GPU
 - AI/Vision Features: Dual ISP, Camera support,
 - Lightweight AI inference
- **GPIO & Expansion:**
 - UNO Compatibility: Compatible with classic Arduino UNO shields
 - Interfaces: Traditional UNO headers, High-speed connectors, Qwiic connector
 - Supports: Sensors, Displays, Cameras, Motor drivers
- **USB & Multimedia:**
 - USB Port: USB-C
 - Supports: Power, Data, Video output, USB peripherals
 - Peripheral Support: Keyboard, Mouse, Camera, Monitor (through dongle)
- **LED Matrix:**
 - Display Type: 8×13 LED Matrix
 - Application: Animations, Status display, Debugging
- **Power Specifications:**
 - USB-C Input: 5V @ 3A
 - VIN Input: 7V – 24V
- **Software Support:**
 - Development Environments: Arduino IDE, Arduino App Lab, Python, Linux terminal
 - Support: Arduino sketches, Linux applications, AI models

2. 2.4" TFT Display

- Display Type: TFT LCD (Thin Film Transistor)
- Display Size: 2.4 Inch
- Display Resolution: 240×320 Pixels (QVGA)

- Display Colors: 65K / 262K Colors
- Controller IC: ILI9341 / ST7789 / HX8347(depends on module)
- Interface Type: SPI / 8-bit Parallel
- Operating Voltage: 3.3V – 5V
- Logic Level: 3.3V TTL
- Backlight Type: LED Backlight
- Touch Support: Optional Resistive Touch
- Viewing Angle: Wide Viewing Angle
- Refresh Rate: Up to 60 Hz
- Display Orientation: Portrait / Landscape
- Pixel Format: RGB565
- Current Consumption: 40mA – 120mA (depends on backlight)

3. Ethernet Hat W5100:

- Module Name: Ethernet Hat / Shield W5100
- Main Controller: WIZnet W5100 Ethernet Controller
- Network Standard: IEEE 802.3 10/100 Ethernet
- Interface Type: SPI (Serial Peripheral Interface)
- SPI Speed: Up to 14 MHz
- Supported Protocols:
 - TCP: Supported
 - UDP: Supported
 - ICMP: Supported
 - IPv4: Supported
 - ARP: Supported
 - IGMP: Supported
 - PPPoE: Supported
- Socket Support: 4 Independent Hardware Sockets
- Internal Buffer Memory: 16 KB
- MAC Address: User Configurable
- PHY Interface: Integrated Ethernet PHY
- RJ45 Connector: Integrated with Link/Activity LEDs
- Operating Voltage: 5V
- Logic Level: 3.3V Internally
- Current Consumption: Approx. 150mA – 180mA
- Communication Interface: SPI
- Additional Features:
 - MicroSD Slot: Available on some shields
 - Reset Controller: Included
 - Status LEDs: Power, Link, Full Duplex, Activity LEDs

- Network Speed: 10 Mbps / 100 Mbps
- Cable Support: CAT5 / CAT5e Ethernet Cable

NS3 (Network Simulator 3)

- **High Fidelity:** It provides highly accurate models of how packets behave. If you write code for a protocol in NS3, it is designed to be easily "ported" to real-world hardware.
- **Modular Architecture:** Everything is a "node." You can plug in different "NetDevices" (like Wi-Fi, LTE, or CSMA) and "Channels" to simulate complex environments.
- **Performance Analysis:** It generates incredibly detailed data. Researchers use it to measure **latency, jitter, packet loss, and energy consumption** for scientific papers.
- **Protocol Development:** Because it is **open-source**, you can modify existing protocols (like TCP or OSPF) or invent entirely new ones to see how they perform.

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